



MANUAL NUMBER: CD32ZZ-080D

DATE: Feb. 20 / 2020

TECHNICAL INSTRUCTION

(Original Instruction)

RDK-408Q ***10K COLD HEAD***

For Bruker

For Service Personnel Only

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RDK-408Q

REVISION CONTROL

Rev.	Date	Description	Prepared	Checked	Approved
A	Oct. 11, 2000	Publication of first edition.	—	—	—
B	Feb. 5, 2010	The Time Counter was added. The Humidity Range was added.	—	—	—
C	Sep. 30, 2013	CE declaration was added.	—	—	—
D	Feb. 20, 2020	CE Declaration was deleted.	T. Shiraishi	T. Matsui	M. Saito

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NOTES TO USER

This Cold Head based on our standard 10K Cold Head Model: RDK-408S.

The specification is same as RDK-408S expect the 1st heat load flange size and Time Counter.

Please refer to the “TECHNICAL INSTRUCTION of RDK-408S” about the instruction for the installation, operation and maintenance of this Cold Head.

1 GENERAL INFORMATION

1-1 SPECIFICATIONS

The specifications of the RDK-408Q Cold Head are summarized in **Table 1.1**

Table 1.1 RDK-408Q COLD HEAD SPECIFICATION

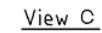
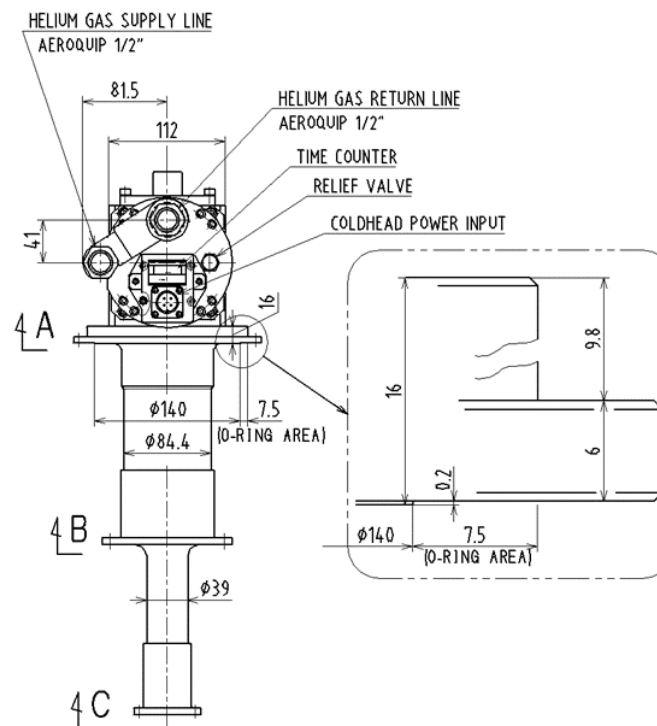
Refrigeration Capacity First Stage Second Stage	30 / 35 W at 45 K (50 / 60 Hz) 5.4 / 6.3 W at 10 K (50 / 60 Hz)
Orientation	Any (Capacity Loss: Max. 15%)
Ambient Temperature Range	5 to 35 deg.C (28 to 35 deg.C with 5% Capacity Loss)
Humidity Range	25 to 85 %RH (without dew)
Helium Gas Pressure for Indoor Used Compressor Unit Static Operating (High Side)* (for Reference) for Outdoor Used Compressor Unit Static Operating (High Side)* (for Reference)	1.45 - 1.50 MPa at 20 deg.C 2.00 - 2.20 MPa --- approx. 1.45 - 1.55 MPa at 20 deg.C 1.90 - 2.10 MPa --- approx.
Pressure Relief Valve Setting	1.86 - 1.96 MPa
Gas Supply Connector Gas Return Connector	1/2-inch Coupling 1/2-inch Coupling
Dimension Width Length Height	180 mm 299 mm 520 mm
Mass	17.2 kg --- approx.

* The operating pressure varies according to the heat load of cold head and temperature around the equipment.

APPENDIX

DRAWINGS

No.	PART NAME
1	RDK-408Q COLD HEAD

[illegible]

RDK-408Q COLD HEAD